



AP Stats Carnival Game Project: GEAR PONG

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What is a Fair Game?

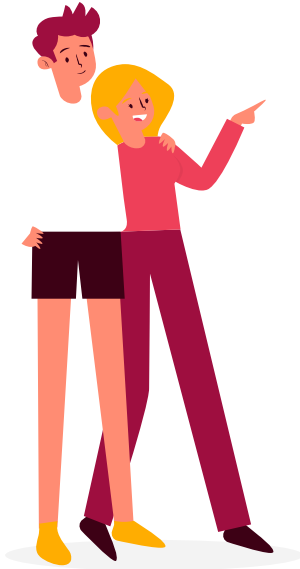
A fair game is a game in which the expected net profit per play equals \$0; this means whenever one plays this game, on average, they will win back the amount that they paid for the game.

On the other hand, most carnivals and amusement parks generate revenue with unfair games, where on average, players will not win back the amount of money they pay for a game (the game has a positive expected net profit).





Game Name: Gear Pong



In this game you place a ball in the top tube, where it falls into a plinko board, and then either directly into a cup or into a box. In the box, the ball will land on a gear, and then fall into a cup; spin the handle to spin the gear! Repeat this with two more balls and total up your points for a prize!

How to Play



1

Drop a ball into the top tube.



2

Watch it fall through the Plinko Board; if it exits through one of the middle two tubes, spin the crank to spin the gear.



3

Collect the ball from the cup it lands in, and use the side door if the ball is in the box. Note how many points you score (black = 0 points, red = 1 point, gold = 3 points)



4

Repeat steps 1-3 two more times. Total up your points across the three balls and (hopefully) collect your prize (5 or more = grand prize, 3 or 4 = regular prize, 2 or less = no prize)!

Parts of the Game

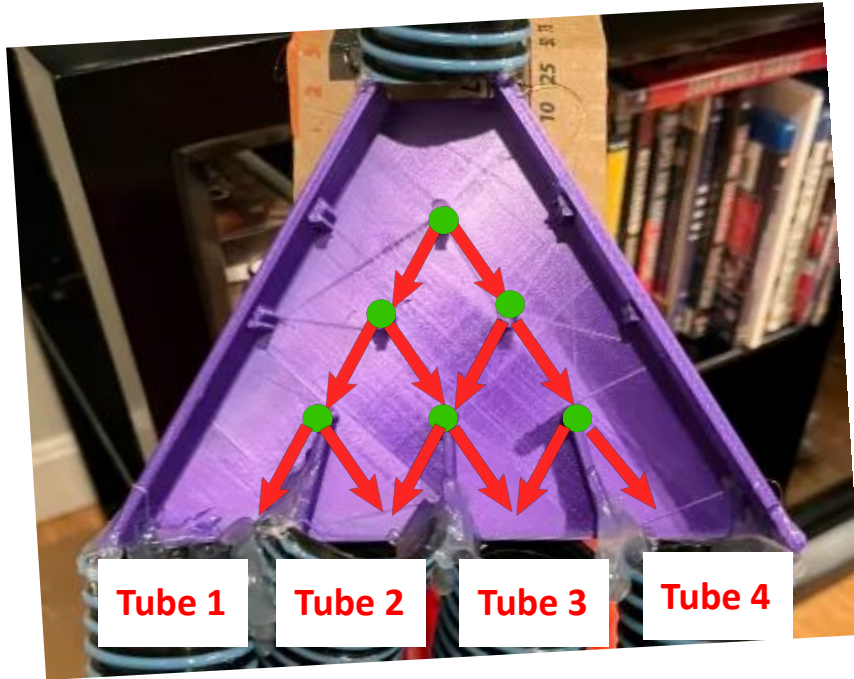
Plinko Board:



Box, with gear and cups inside:



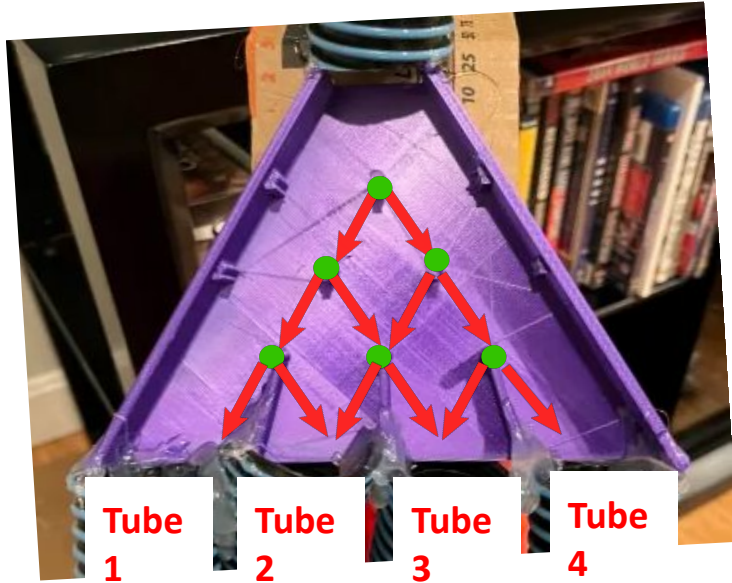
Probability Calculations



When a ball enters the plinko board, it hits the top peg, and then exits through a tube following one of the paths shown by the red arrows. In total, it hits 3 pegs, and after it hits each peg, it bounces to the left or to the right, each with a 50% chance (because all the pegs are equally spaced from each other).

The tube which the ball travels down is determined by the number of pegs which the ball bounces right after hitting; 0 right bounces mean tube 1, 1 right bounce means tube 2, 2 right bounces means tube 3, and 3 right bounces means tube 4.

Probability Calculations



Now, define the random variable X so that X = the number of right bounces in 3 trials. Because the ball has a 50% chance of bouncing either way after hitting each peg, and 3 pegs are hit, X can be modeled by $\text{Binom}(n = 3, p = 0.5)$.

Then, the probabilities of the ball falling into each tube are:

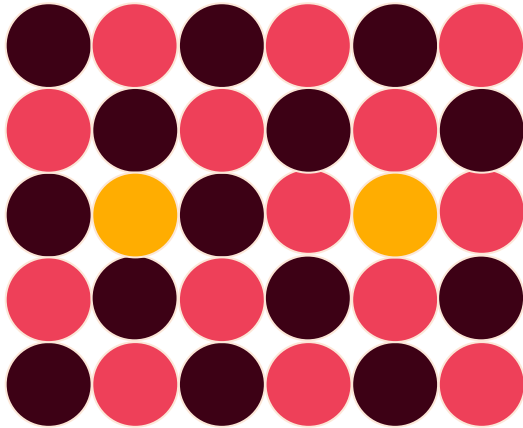
$$P(\text{tube 1}) = P(X = 0) = \text{binompdf}(n = 3, p = 0.5, X = 0) = 0.125$$

$$P(\text{tube 2}) = P(X = 1) = \text{binompdf}(n = 3, p = 0.5, X = 1) = 0.375$$

$$P(\text{tube 3}) = P(X = 2) = \text{binompdf}(n = 3, p = 0.5, X = 2) = 0.375$$

$$P(\text{tube 4}) = P(X = 3) = \text{binompdf}(n = 3, p = 0.5, X = 3) = 0.125$$

Probability Calculations



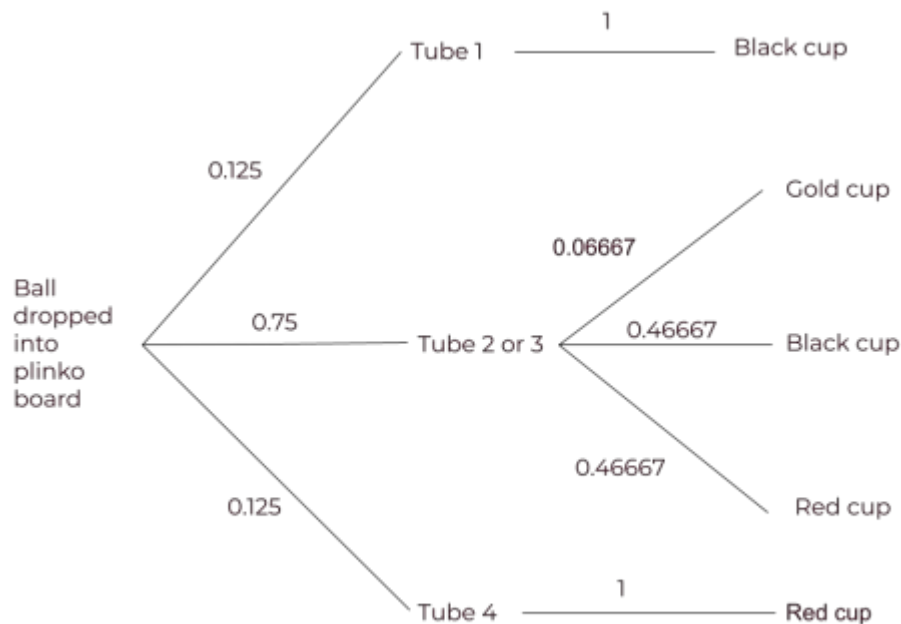
If the ball gets in the box (which happens with probability $0.375+0.375=0.75$), the chance of getting into one of 30 cups is random due to the mechanics of the game (gear spinning and highly bounce ball). There are with 14 red cups, 14 black cups and 2 gold cups so:

$$P(\text{Black cup}) = 14/30 = 0.46667$$

$$P(\text{Red cup}) = 14/30 = 0.46667$$

$$P(\text{Gold cup}) = 2/30 = 0.06667$$

Probability Calculations



So, considering all possibilities, for each ball, the probability of getting each color is as follows:

- $P(\text{gold cup}) = P(\text{tube 2 or 3}) \cdot P(\text{gold cup} \mid \text{tube 2 or 3}) = 0.75 \cdot 0.06667 = 0.05$
- $P(\text{red cup}) = P(\text{tube 2 or 3}) \cdot P(\text{red cup} \mid \text{tube 2 or 3}) + P(\text{tube 4}) \cdot P(\text{red cup} \mid \text{tube 4}) = 0.75 \cdot 0.46667 + 0.125 \cdot 1 = 0.475$
- $P(\text{black cup}) = 1 - P(\text{gold cup}) - P(\text{red cup}) = 1 - 0.05 - 0.475 = 0.475$



Probability Calculations

For 3 balls, here are all outcomes which get a grand prize (5 or more points):

Outcome	Points	# of Orderings (gold = G, red = R, black = B)	Probability of occurring
3 golds	9	1 - only GGG	$1 \times (0.05)^3 = 0.000125$
2 golds, 1 red	7	3 - GGR, GRG, RGG	$3 \times (0.05)^2 \times (0.475) = 0.0035625$
2 golds, 1 black	6	3 - GGB, GBG, BGG	$3 \times (0.05)^2 \times (0.475) = 0.0035625$
1 gold, 2 reds	5	3 - GRR, RGR, RRG	$3 \times (0.05) \times (0.475)^2 = 0.03384375$

Therefore $P(\text{grand prize}) = 0.000125 + 0.0035625 + 0.0035625 + 0.03384375 = \boxed{0.0411}$



Probability Calculations

Here are all of the outcomes for a regular prize after 3 balls (3 or 4 points):

Outcome	Points	# of Orderings	Probability
1 gold, 1 red, 1 black	4	6 - GRB, GBR, RGB, RBG, BGR, BRG	$6 \times (0.5) \times (0.475) \times (0.475) = 0.0676875$
1 gold, 2 blacks	3	3 - GBB, BGB, BBG	$3 \times (0.05) \times (0.475)^2 = 0.03384375$
3 reds	3	1 - RRR	$1 \times (0.475)^3 = 0.107171875$

So, $P(\text{regular prize}) = 0.0676875 + 0.03384375 + 0.107171875 = \boxed{0.2087}$

$P(\text{no prize}) = 1 - P(\text{grand prize}) - P(\text{regular prize}) = 1 - 0.0411 - 0.2087 = \boxed{0.7502}$

Prizes



Regular - 3 or 4 pts

Win a small fidget toy (worth \$0.18)



Grand - 5 pts

Win a large teddy bear (worth \$28.99)



Expected Value

Let C be a random variable denoting the cost (in dollars) of one play. Using the prize values, the following is a probability model for C :

c	$P(C=c)$
28.99	0.0411
0.18	0.2087
0	0.7502

The expected cost of one play is

$$E(C) = \sum c_i P(C = c_i) = 28.99 \cdot 0.0411 + 0.18 \cdot 0.2087 + 0 \cdot 0.7502 = \$1.23.$$

We will charge \$3 per play.

Now, assume 100 people play the game. The expected cost is

$$E(C_1 + C_2 + \cdots + C_{100}) = E(C_1) + E(C_2) + \cdots + E(C_{100}) = 100 \cdot 1.23 = \boxed{\$123}$$

Because the gross profit of one play is \$3.00, the expected gross profit of 100 players is

$$100 \cdot 3.00 = \boxed{\$300}.$$

So, the expected net profit is

$$\begin{aligned} E(\text{net profit after 100 players}) &= E(\text{gross profit of 100 players} - \text{cost of 100 players}) \\ &= E(\text{gross profit of 100 players}) - E(C_1 + C_2 + \cdots + C_{100}) \\ &= 300 - 123 \\ &= \boxed{\$177} \end{aligned}$$

The average profit per game is

$$\begin{aligned} E(\text{profit of one game}) &= E(\text{gross profit of one game} - \text{cost of one game}) \\ &= E(\text{gross profit of one game}) - E(C) \\ &= 3.00 - 1.23 \\ &= \boxed{\$1.77} \end{aligned}$$

Results from Carnival Day

During Carnival Day, 12 people played our game. The results are listed in the table to the right. Across the 12 players, 1 won a grand prize, 2 won regular prizes, and 9 won no prizes. Since each player “paid” \$3 to play, our actual net profit was

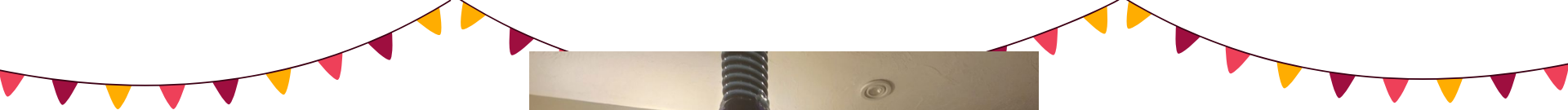
$$15 \cdot 3 - (28.99 + 2 \cdot 0.18) = \$15.65$$

However, our expected net profit was

$$\begin{aligned} E(\text{net profit after 15 players}) &= E(\text{gross profit of 15 players} - \text{cost of 15 players}) \\ &= 15 \cdot 3 - E(C_1 + C_2 + \cdots + C_{15}) \\ &= 45 - 15 \cdot 1.23 \\ &= \$26.55 \end{aligned}$$

Our actual net profit was quite different from our expected net profit, although we were still able to make a decent amount of money (average of \$1.04 of net profit per person). This difference is likely due to the fact that one player managed to win the grand prize of a giant teddy bear, worth \$28.99.

Player Number	Points Scored	Prize
1	2	None
2	1	None
3 (Giang)	5	Grand
4	2	None
5 (Tanush)	3	Regular
6	1	None
7	0	None
8	2	None
9	2	None
10	2	None
11	1	None
12 (Mr. Burns)	3	Regular
13	0	None
14	2	None
15	1	None





OVERALL REFLECTION



WAS IT A SCAM?

For most people, our game was a scam as they were paying \$3.00 to play and were not able to win anything. However, for one lucky person this game was certainly not a scam as they won the grand prize.



WAS IT FAIR?

No, a fair game needs an expected net profit of 0, meaning that a player would on average not lose any money. Our game had an expected net profit of \$1.77, making it not fair.



HOW DID WE DO IT?

We made the pinball board and gears very precisely using TJ's 3d printer. We also randomized the probability of the ball landing in each cup in the box by using a very bounce ball and small, equally-sized cups.



LAST

THOUGHTS?

We learned that probability is a useful tool for estimating results, but doesn't always turn out exactly as expected.